

Hafiz M. Hassan

AI Engineer · Data Scientist · Full-Stack Developer · Vibe Coder

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PROFESSIONAL SUMMARY

Driven Data Science student at FAST NUCES — Pakistan's most prestigious CS institution — operating at the intersection of AI, full-stack web engineering, and applied machine learning. Co-founded QuickSite, a production web agency, and is currently conducting active research on NLP2Shell — a fine-tuned language model that translates natural language to shell commands. Keeping up with the market by developing an efficient workflow of AI tools for Vide Coding, providing with better code in lesser costs. Open to challenging roles in AI engineering, data science, and full-stack development.

EDUCATION

BS Data Science

2024 – Present

FAST NUCES, Lahore · Pakistan's #1-ranked CS institution

- Coursework: Data Structures & Algorithms, Machine Learning, Databases, OOP, Digital Logic Design
- Actively conducting independent AI/NLP research (NLP2Shell — see Projects)

Intermediate in Computer Science (ICS)

2022 – 2024

Government College University (GCU), Lahore · Premier national institution

- High-merit admission; strong foundation in mathematics, programming logic, and computational thinking

Matriculation (Science)

2020 – 2022

UGI, Lahore

- Distinction-level academics; initiated programming interest

EXPERIENCE

Machine Learning Intern

Jul 2026 – Cont.

FlyRank AI – Chicago, Illinois, USA · Remote

- Completed a very sophisticated “learning by practical work” cohort track.
- Performed various ML tasks on real world data and completed a capstone in ML based SEO.

Data Analysis and ML Engineer Inter

Jun 2026 – Jul 2026

GamiTron – Lahore

- Applied data analysis in the company's core infrastructure data.
- Trained various ML models for different tasks, expanded my domain knowledge in ML for game dev.

Programming Fundamentals — Teacher Assistance

2026 – Present

FAST NUCES · 25 Batch PF's Summer TA

- Helped students with for better understanding of Fundamental concepts of Programming

- Provided Assistance with Assignments and Quizzes preparation, conductance and checking

Co-Founder & Developer — QuickSite

2024 – Present

QuickSite Web Agency · [quick-site-00.vercel.app](#)

- Co-founded a web agency delivering fast, production-ready one-page sites for founders and businesses
- Engineered automated build + Vercel deploy pipeline achieving 100% deployment automation

Junior Game Development Trainer

Aug 2022 – Jul 2023

GCUL – FSDS (Freelancing & Skill Development Society), Lahore

- Assisted senior trainer at GCUL's leading tech society, delivering structured game-dev.
- Trained Students in game development by online sessions and GCRs

ACTIVE RESEARCH

NLP2Shell — Natural Language to Shell Command Execution

2024 – Present

Fine-tuned LLM research project · [github.com/HassanNawaz14/nlp2shell](#)

- Research question: *What if we could bypass the GUI layer with even more efficient and easier to interact layer of NLP2Shell translator between the OS and the User?*
- Fine-tuned Qwen2.5-0.5B on 26,000 NL→Bash pairs using LoRA (r=16) + 4-bit NF4 quantisation on a Kaggle T4 GPU
- Built a full inference pipeline: Whisper STT → LLM predict → safety layer → subprocess execution
- Designed a hardcoded safety blacklist preventing destructive commands (rm -rf, etc.); mandatory confirmation loop
- Phases completed: data prep, fine-tuning, inference pipeline; CLI interface active; evaluation next

PROJECTS

🔗 ZeroAbstractionNet Python · C · ASM · Deep Learning · ML

- A fully hand written MLP (ANN) written without any libraries (not even NumPy).
- Matmul (in FP, BP) written in Python, C and ASM for performance optimization.

🔗 Sahulat Web Next.js · Data Scrapping · Node.js · Express

- A multi-purpose bills, outages, budget tracking service.
- Mobile friendly, SEO optimized frontend, maximized user friendly useful features.

🔗 TravelBuddy React · Node.js · Express · SQL Server

- Full-stack travel planning platform with buddy matching algorithm for expense sharing.
- Features: trip planner, budget estimator, Vloggers Hub, responsive modern UI.

🔗 CADS-Bridge React · Node.js · Express · SQL Server

- Real-time collaboration platform bridging CA and Data Science teams within firms.
- Role-based dashboards, KPI tracking, audit logging, annotation engine, automated conflict detection.

🔗 Sentiment Predictor Python · TF-IDF · Scikit-Learn · NLP · ML

- End-to-end sentiment analysis pipeline trained on textual reviews with multiple ML model comparisons.

🔗 Fashion Image Classifier Python · YOLO · CV · CNN · ML

- An image classifier that classifies fashion products using YOLO for object detection and custom CNN for classification.

🔗 CPU Storm OS · Ethical Hacking · Algorithms

- A simulation of an actual DDoS – Attack, simulating attack by making CPU stave by 100% usage.

🔗 RecommendationX ML · JS · Node.js

- A movie recommendation system through the Cosine Similarly Algorithm that recommends on the basis of other users, not you own history.

🔗 Brick Breaker + Space Shooter Assembly x86 · Game Dev

- Classic games built in raw 8086 Assembly using direct video memory access and custom interrupt handling.

🔗 GearPro · LudoWeb · Solar App · LibMS · ReLive HTML · CSS · JS · C++

- Additional portfolio: movie recommender, car specs database, 4-player Ludo, solar recommendation engine, Library Management System, OOP social network.

SKILLS & TECHNOLOGIES

Languages	Python · C · C++ · C# · JavaScript · Assembly x86 · SQL
Data Science	Data Analysis · Data Warehousing · Pandas · NumPy · Ad. Statistics
AI / ML	Keras/TensorFlow · Scikit-Learn · NLP · Deep Learning · LLMs · Gen AI
Web / Full-Stack	React · Next.js · Flask · Node.js · Express · HTML5 · CSS3 · SQL Server · Vercel
Game Dev	Unity · Assembly x86 · C#
Tools	Git · GitHub · VS Code · Jupyter · Kaggle · MS Office · Ubuntu · APIs · MS office

CERTIFICATIONS & CONTINUED LEARNING

Anthropic	Claude 101 · Claude Code 101 · Claude Code in Action · Agent Skills (Certificates)
Google	Introduction to Generative AI · Introduction to Large Language Models (Certificates)
Kaggle	Pandas · Data Cleaning (Certificates)
Microsoft	Azure Data Fundamentals (in progress) · Multiple Microsoft Learn skill badges

COURSES & SELF LEARNING

MIT OpenCourseWare	18.04: Linear Algebra (Gilbert Strang) · How To Speak (PHW) · 6.034: Artificial Intelligence (Patrick H. Winston) · 6.046J Introduction to Algorithms
Harvard	CS50: Introduction to the intellectual enterprise of Computer Science
Stanford	CS231n: Deep Learning for Computer Vision, CS244n: NLP
Oxford	An AI Stack
CampusX	100 Days of ML · NLP · 100 Days of Deep Learning · Generative AI
FreeCodeCamp	Selenium & Browser Automation · HTML, CSS, JavaScript, Node.js, Express
Net Ninja	React · React Router · Node.js · Git

ONLINE PRESENCE

 Kaggle	 LeetCode	 Fiverr
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Recommendations Available on Request